

Mogal ashig _192312351

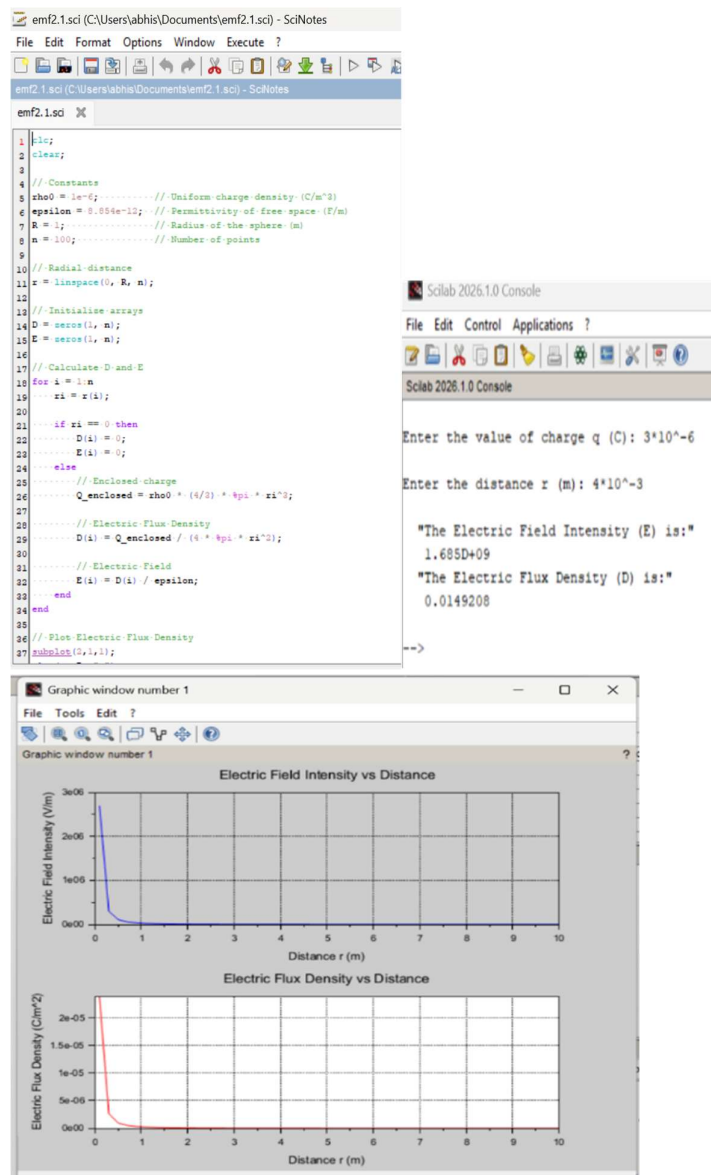
Emf_eca0502

Exp. No: 2 Determination and plotting of Electric field and Electric flux density in sphere full of

Aim: To write program for electric field and electric flux density in sphere using SCILAB.

Software required:

- SCILAB version 6.0.1



Given:

$$q = 3 \times 10^{-6}$$

$$r = 4 \times 10^{-3} \text{ m}$$

Formula

$$E = \frac{q}{4\pi\epsilon_0 r^2}$$

$$r^2 = (4 \times 10^{-3})^2 = 16 \times 10^{-6} = 1.6 \times 10^{-5} \text{ m}^2$$

$$4\pi\epsilon_0 r^2 = (4)(3.1416)(8.854 \times 10^{-12})(1.6 \times 10^{-5})$$
$$= 1.7803 \times 10^{-15}$$

$$E = \frac{3 \times 10^{-6}}{1.7803 \times 10^{-15}} = 1.685 \times 10^9 \text{ V/m}$$

$\frac{dV}{dr}$

$$E = 1.576 \times 10^9 \text{ V/m}$$

Electric Flux Density (D)

Formula

$$D = \epsilon_0 E$$

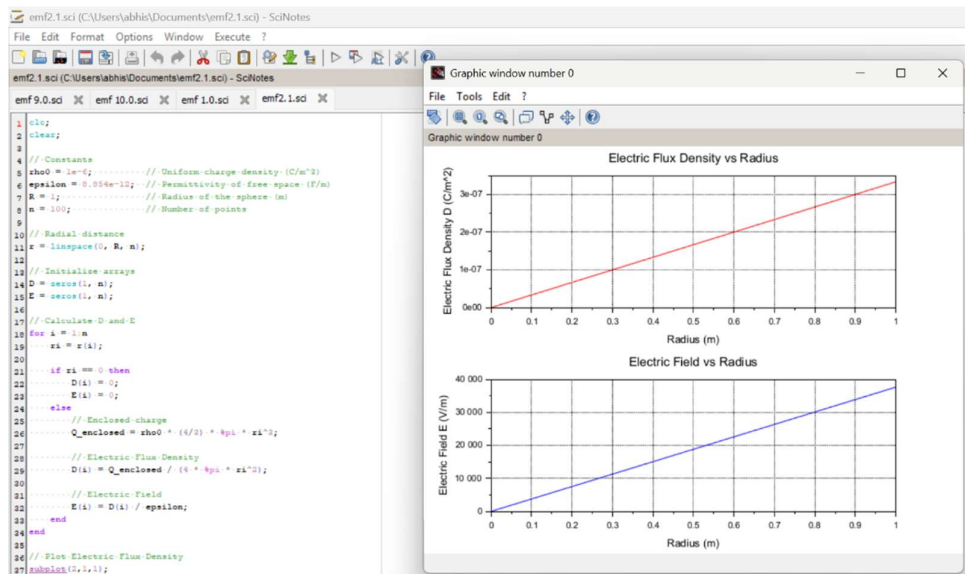
$$D = (8.854 \times 10^{-12})(1.685 \times 10^9)$$

$$D = 1.492 \times 10^{-2} \text{ C/m}^2$$

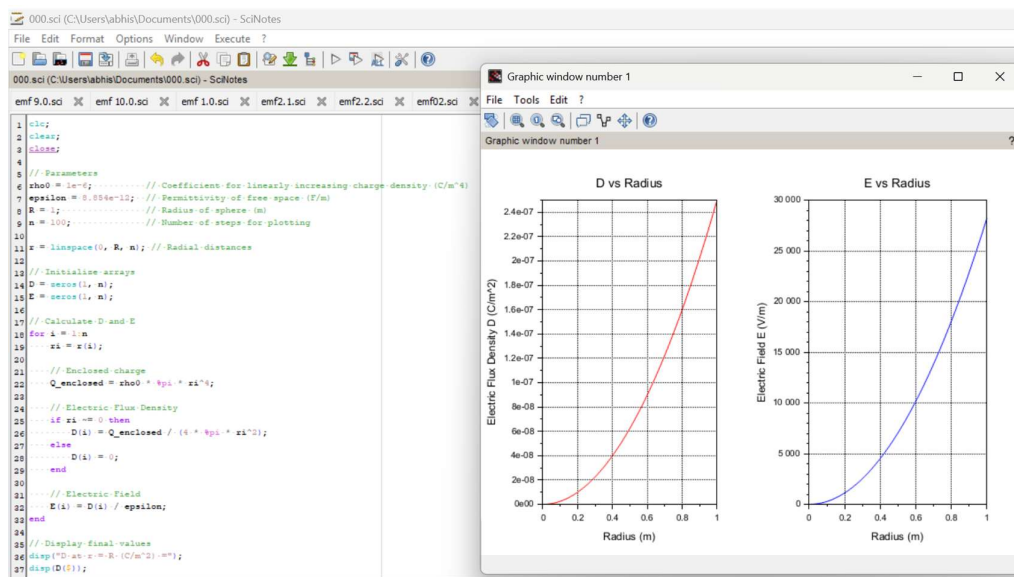
$$D = 0.0015951$$

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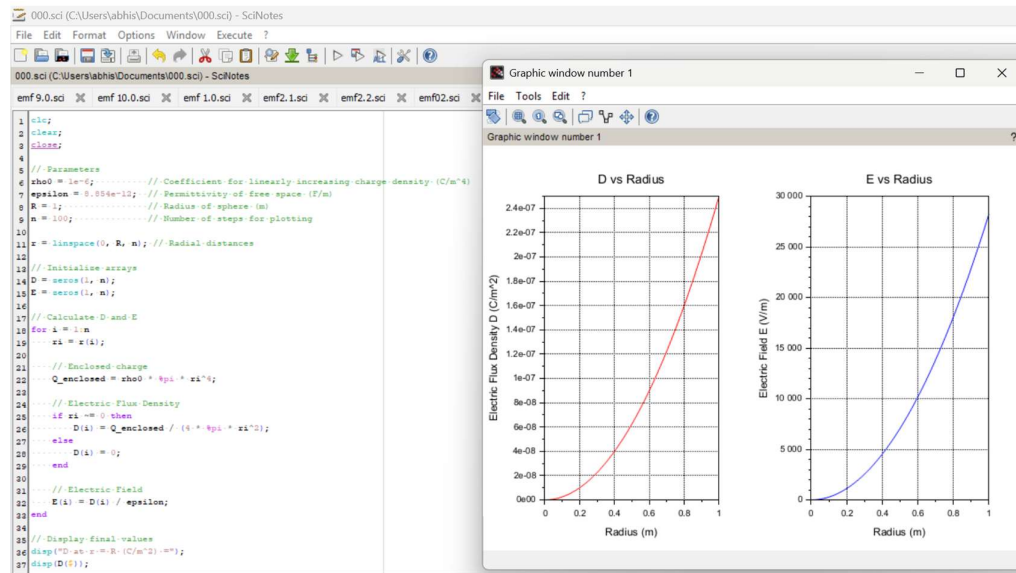
Case 1: Derive and plot the electric field (E) and electric flux density (D) inside a uniformly charged sphere with charge density ρ_0 . What are the values of E and D at the center and at the surface of the sphere?



Case 2: For a sphere with a linearly increasing charge density $\rho_v = \rho_0 r$, derive the expressions for the electric field (E) and electric flux density (D). Plot their variations inside the sphere and analyze the trends.



Case 3: A sphere of radius $R=2$ m has a uniform charge density $\rho=3 \times 10^{-6}$ C/m³.
 Calculate and plot the electric flux density $D(r)$ and electric field $E(r)$ as a function of radial distance r , ranging from 0 to 4 meters.
 2. Discuss the behaviour of the electric field inside and outside the sphere.



Result : Thus the program was verified for electric field and electric flux density in sphere using SCILAB was successfully.